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Takenaka Corporation Promotes Digital Transformation by Building a Digital Construction Solution on AWS

Takenaka Corporation Promotes Digital Transformation by Building a Digital Construction Solution on AWS and Improved Use of the Construction Business Data

[Overview](#) | [Opportunity](#) | [Solution](#) | [Architecture](#) | [Outcome](#) | [AWS Services Used](#)

100+ applications
developed on AWS

19
to c  Hi, I can connect you with an AWS representative or answer questions you have on AWS.

Expanded data use

D  Need more info? Highlight any text to get an explanation generated with AWS generative AI.

Since its establishment in 1610, [Takenaka Corporation](#) has worked on many architectural projects. The company built a digital construction solution, which was the foundation for its digital transformation, on Amazon Web Services (AWS), and this solution was launched in November 2021. The digital construction solution efficiently develops applications that are necessary for digital transformation by implementing data collection, artificial intelligence (AI), business intelligence (BI), and Internet of Things (IoT) functions. Takenaka Corporation aims to improve productivity drastically through digitizing all operations, including construction sites, and add value through advanced data use.



Opportunity | Improving Data Management in Construction Projects

Takenaka Corporation aims for sustainable construction and community development, with “being involved in community development and creating new value as a group, globally” as its 2025 growth strategy. One of the efforts to implement this strategy is digital transformation. The company’s digital strategy is to align business departments with the digital office, streamline all operations with digital technology, and promote digital transformation across divisions and corporate barriers.

“In the construction industry, productivity has remained almost flat for the past 20 years,” says Dr. Hideo Kitahara, Group Leader for Futuristic Digital Technology Group of Digital Division at Takenaka Corporation. “The aging of skilled workers has also progressed, and the shortage of human resources has become serious.” Additionally, the revised Labor Standards Law is being enforced in the construction industry in 2024. This law limits overtime work to 45 hours per month per employee. Considering these factors, improving work efficiency and productivity through digitalization is essential for the company to grow.

The core of the digital strategy is a digital construction solution, which aggregates all business data and makes it usable. The aim is to promote new value creation by integrating and managing all business data as sales, design, production preparation, construction, maintenance, personnel, and accounting.



“Until now, systems have been siloed for each division, such as sales, design, and management, and data use for construction projects has been closed on a business basis,” says Yoko Matsumoto, Group Leader for Data Management Group of Digital Division at Takenaka Corporation. “Therefore, we have built a digital construction solution as a common foundation that facilitates data use across the entire company.”

Solution | Building a Digital Construction Solution with Data Collection, AI, BI, and IoT Functions

Takenaka Corporation began considering a digital construction solution in May 2019, then examined infrastructure based on the cloud and adopted AWS after comparing multiple providers.

“We are proceeding with the migration of the entire group’s business system from on premises to AWS around 2025, and we have decided to adopt it from the viewpoint of both a cloud lift and a digital construction solution,” says Shinichi Misato, Chief Solution Creator for Futuristic Digital Technology Group of Digital Division at Takenaka Corporation. “We adopted AWS after comprehensively evaluating the ease of migration of existing systems, market share, sustainability as a cloud service, and the range of advanced services required for digital transformation, such as AI, IoT, and big data.”

Takenaka Corporation started to build the solution in April 2020. After defining requirements, building infrastructure, and collecting data, the company launched the solution in November 2021. It consists of two parts: a data foundation and a suite of applications for digital transformation.

The data foundation is for managing and analyzing all business data. For data storage and collection, the solution uses [Amazon Simple Storage Service](#) (Amazon S3) and [Amazon Relational Database Service](#) (Amazon RDS), and [AWS Glue](#). To run AI and machine learning (ML) models for data analysis, the solution uses [Amazon SageMaker](#). To run the business intelligence workload, the solution uses [Amazon Redshift](#) in addition to Takenaka Corporation’s existing BI tools. It also uses [Amazon Kinesis](#) to provide Internet of Things function, connecting collected data and the AI model.

“For the data infrastructure, we actively use AWS managed services so that we can respond flexibly to changes in data volume and addition of functions,” says Misato.



The second part—the application suite—is for data use. The digital office collaborated with the business departments to proceed with development on a trial basis, and part of the suite was released around September 2020. “The first application we created visualized the schedule and performance of skilled workers’ labor hours in order to create work plans,” says Matsumoto. “Between September 2020 and November 2022, we released about 25 applications with different functions, such as visualizing work styles based on PC logs.”

One such application developed by Takenaka Corporation is the Workforce Loading Predictor, which predicts and simulates the number of personnel required for construction management. The application learns what type of construction is required and the progress of a construction project, then predicts the required human resources. Workforce predictions, which were previously done by each branch office using spreadsheets, can now be calculated using unified standards.

“The Workforce Loading Predictor uses Amazon SageMaker for ML to create the predictive model based on data from past performance,” says Toshiaki Yamanishi from Futuristic Digital Technology Group of Digital Division at Takenaka Corporation. “The application applies the created model to an early-stage construction to predict and visualize the number of required personnel with BI. We took security into consideration when connecting the data collection component and the AI component.”

As the company built the digital construction solution, AWS teams provided support, such as design reviews and improvement proposals through regular meetings and individual questions. Among the types of support, AWS Training and Certification was most effective. “More than half of the project members hadn’t had experience using AWS, but the project progressed smoothly as a result of raising their level through training,” says Kitahara.

Outcome | Expanding Data Use for Takenaka Corporation

Takenaka Corporation’s data use is gradually expanding. The digital office is still calling on business departments to use data while expanding stored data; expanding AI, BI, and IoT solutions; and developing applications for digital transformation.

“Using AWS services, we built a digital construction solution that facilitates data use across the entire company. That means we have the foundation for improving productivity at construction sites,” says Kitahara. “We mainly use data from business systems so far. We will gradually accumulate document files, video and audio data, construction site data, and more through the solution and use the collected data for various purposes. We aim to develop a total of over 100 applications by the end of the fiscal year of 2023.”

Furthermore, the company is looking to build a digital twin that combines the digital construction solution with a real-world construction site. For this project, Takenaka Corporation is considering using [AWS IoT TwinMaker](#), which makes it simpler for developers to create digital twins of real-world systems.

“To achieve the 2030 milestone with our digital strategy, the company will link 3D building information modeling data with construction process progress data, environmental data, worker data, and more, and simulate in advance to resolve process delays, improve quality, and verify safety,” says Kitahara.

Takenaka Corporation’s vision is to deepen cooperation with smart building-management solutions and construction robot solutions developed in-house and to provide new construction and town planning services while planning cooperation with local community data solutions.

About Takenaka Corporation



Since it was founded in 1610, Takenaka Corporation has worked on many buildings that have become landmarks with bearing the master builder spirit in mind. Based on the management philosophy of “contribute to society by passing on the best works to future generations,” the company is involved in the construction of buildings that accurately respond to the needs of the times. Takenaka Corporation aims to solve social issues and achieve rich, safe, and secure urban development through its comprehensive capabilities.

AWS Services Used

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

[Learn more »](#)

Amazon RDS

Amazon Relational Database Service (Amazon RDS) is a collection of managed services that makes it simple to set up, operate, and scale databases in the cloud.

[Learn more »](#)

AWS Glue

AWS Glue is a serverless data integration service that makes data preparation simpler, faster, and cheaper.

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Amazon SageMaker

Amazon SageMaker is a fully managed service that brings together a broad set of tools to performance, low-cost machine learning (ML) for any use case.



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